

SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Enabling resource-efficient, sub-1GB
inference on mobile and edge
devices

Large VLMs Are Powerful but Impractical for Edge Deployment

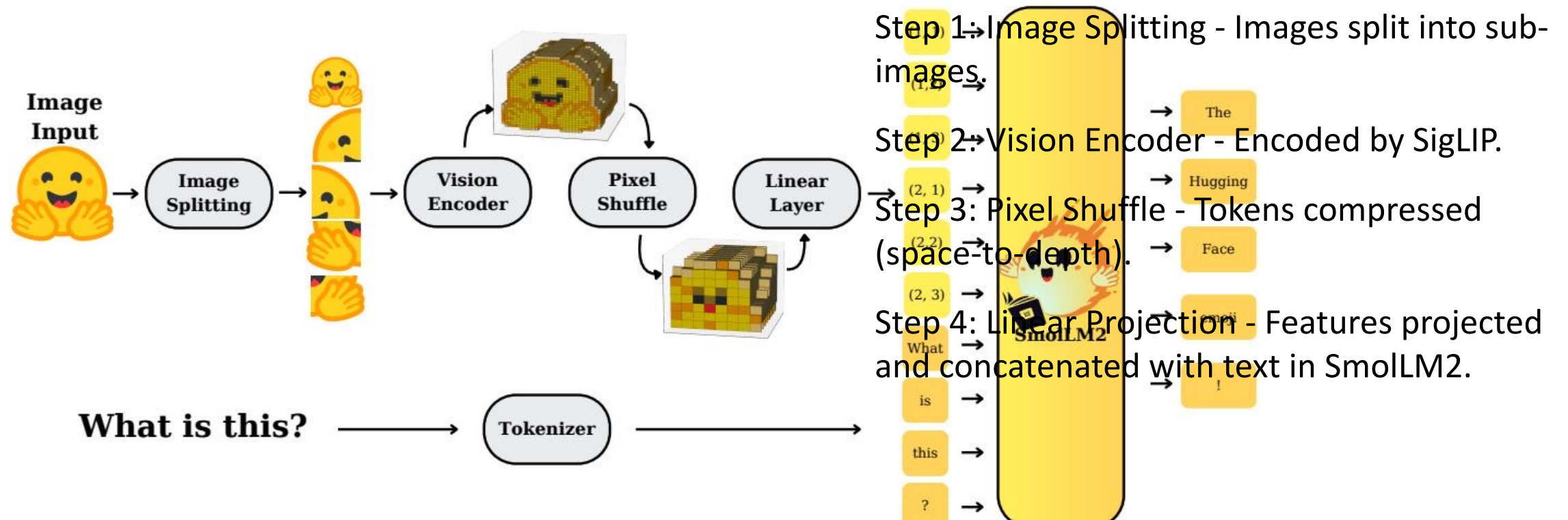
State-of-the-Art Limitations: SOTA Vision-Language Models (VLMs) require massive compute, limiting mobile/edge viability.

Memory Footprint: Leading open models like Molmo-A1B-7B require ~27.7 GB of GPU RAM.

Parameter Bloat: Models like Idefics-80B scale parameters at the cost of accessibility.

The Goal: Achieve competitive multimodal understanding with sub-1GB memory constraints and high tokens-per-second throughput without cloud dependency.

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

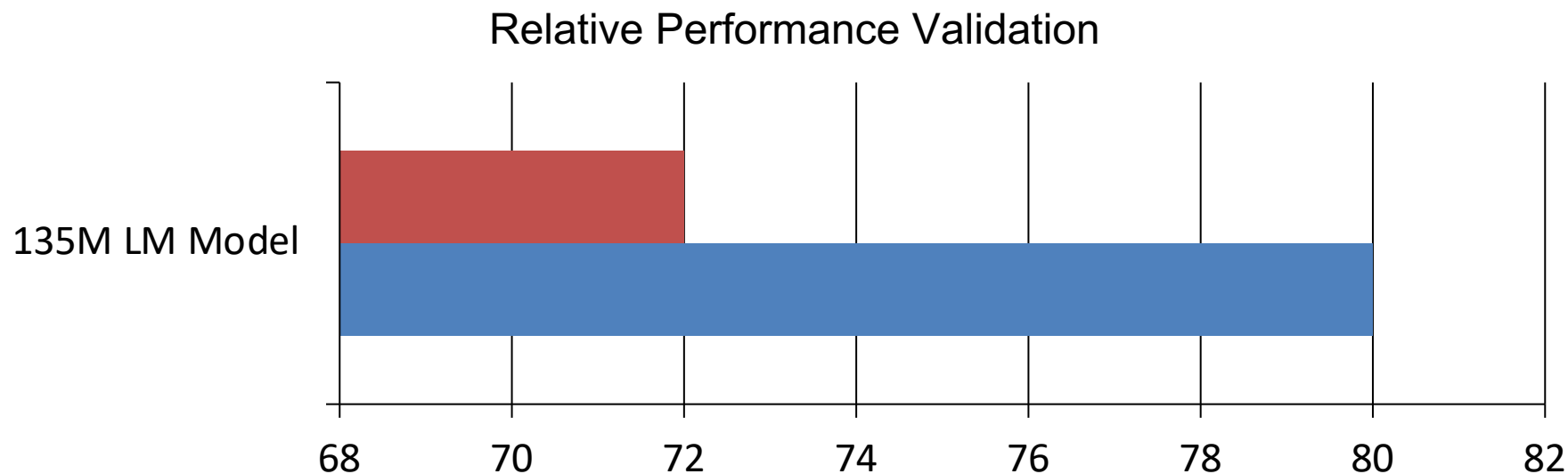


Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Balanced Encoder-LM compute is critical for compact VLMs.

A 93M SigLIP-B/16 vision encoder paired with a 135M LM outperforms a 428M SigLIP-SO400M with the same LM.

Larger encoders (428M) only justify their 10% parameter overhead when the Language Model scales to 1.7B parameters or more.

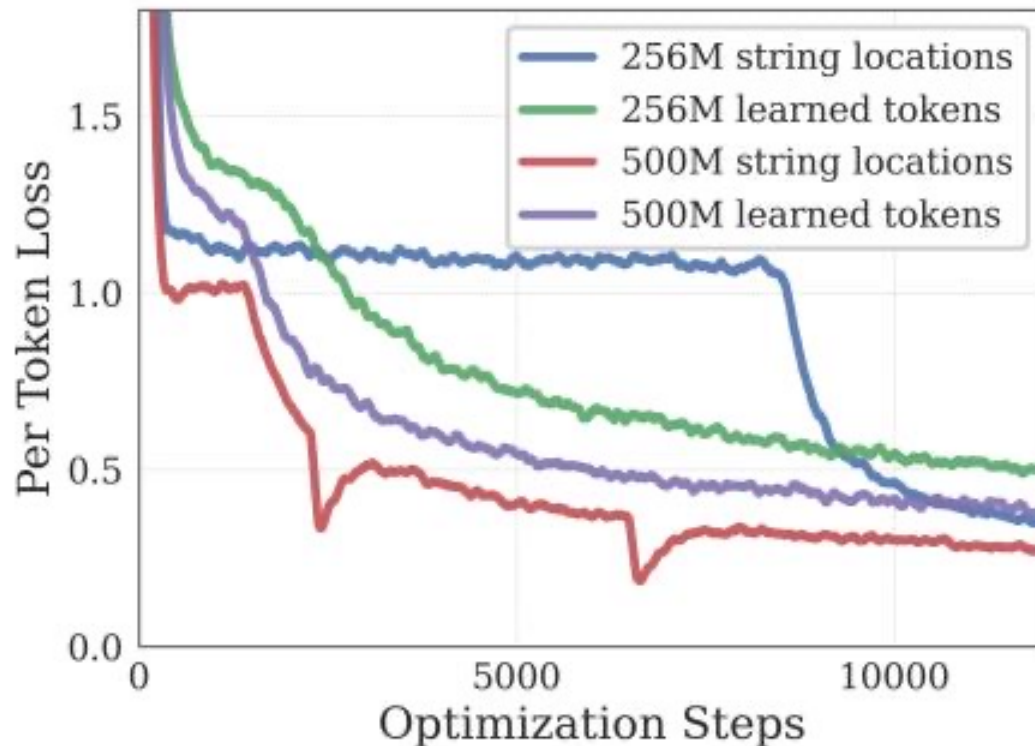


Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Mechanism: Pixel shuffle rearranges spatial features into channels, reducing tokens by r^2 .

Impact: Aggressive compression (r=4) yields a 16× token reduction, easing attention overhead.

Finding: Small VLMs strongly benefit from r=4 to improve long-context modeling, whereas larger models typically use r=2.



Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2: Performance improves significantly when extending context length from 2K to 16K.

Strategy:

- Increased the RoPE (Rotary Position Embedding) base from 10K to 273K.
- Fine-tuned the models on mixed long/short-context data.

Model Specifications:

- SmolVLM-2.2B: Utilizes the full 16K token context.
- SmolVLM-256M & 500M: Utilize 8K token context.

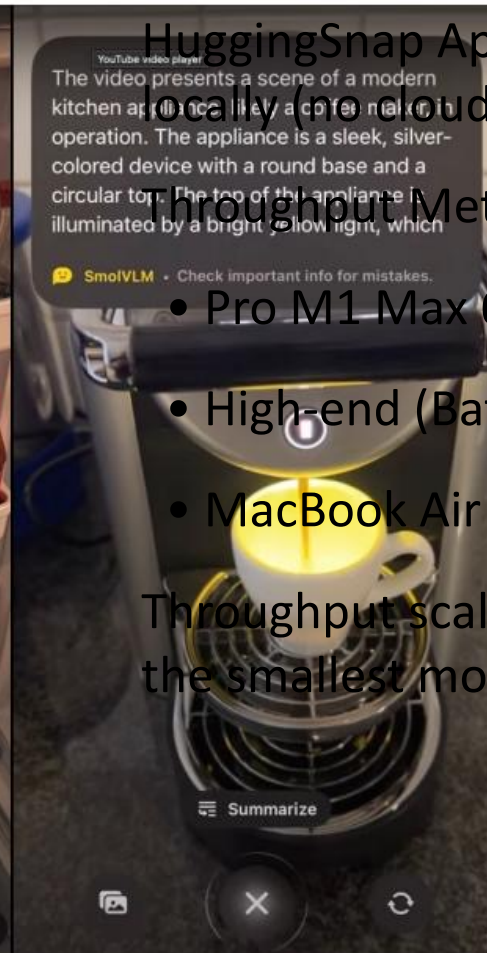
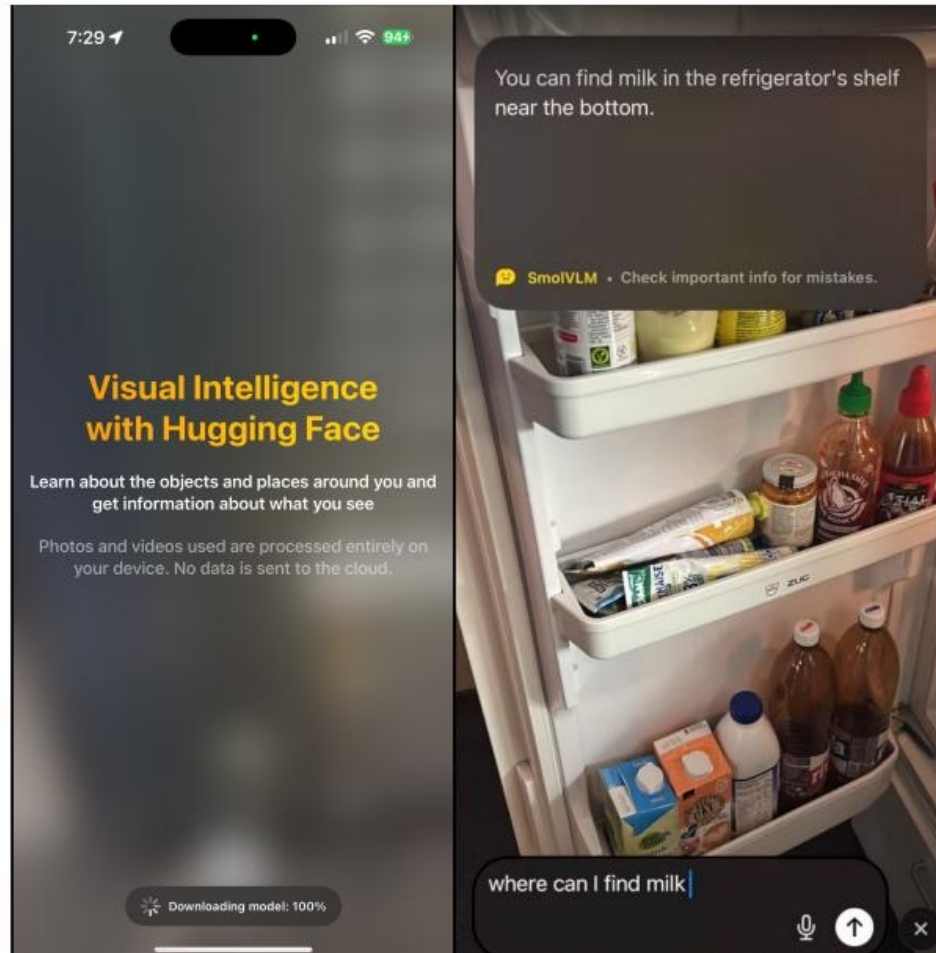
SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Metric	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Avg Score	44.0%	51.0%	59.8%	Molmo-A1B-7B / InternVL2-2B
GPU RAM Usage	0.8 GB	1.2 GB	4.9 GB	27.7 GB (Molmo-A1B-7B)
Scale Note	Surpasses 300x larger Idefics-80B	-	-	-

Benchmark details: Averages taken across 13 benchmarks spanning Single-Image, Multi-task, and Video capabilities.

Key Insight: Substantial performance achieved at a fraction of the hardware requirements.

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

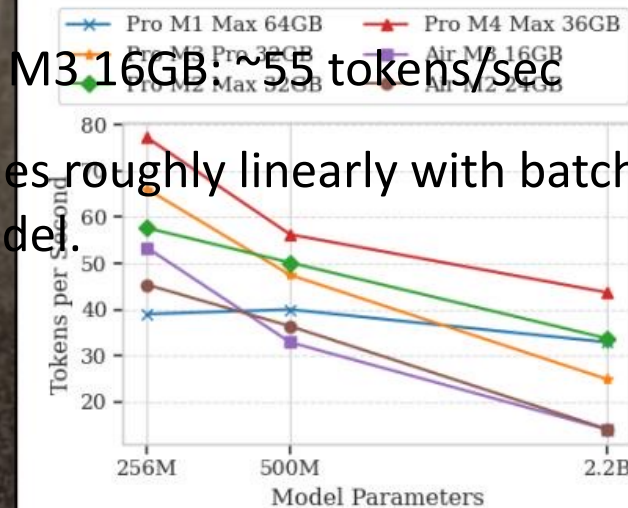
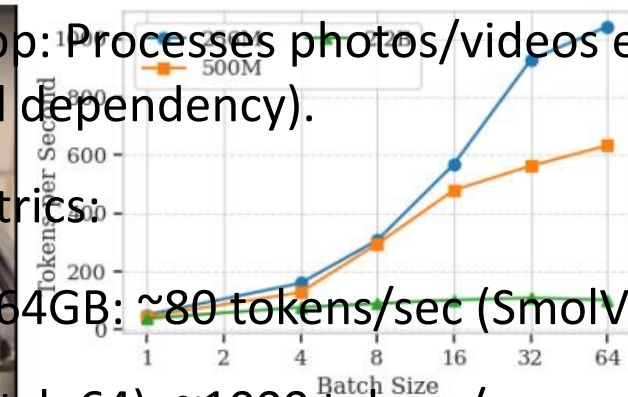


HuggingSnap App: Processes photos/videos entirely locally (no cloud dependency).

Throughput Metrics:

- Pro M1 Max 64GB: ~80 tokens/sec (SmoVLM-256M)
- High-end (Batch 64): ~1000 tokens/sec
- MacBook Air M3 16GB: ~55 tokens/sec

Throughput scales roughly linearly with batch size for the smallest model.



Key Takeaways: Design Principles for Efficient Multimodal Models

1. Balance Encoder-LM Compute: Small LMs demand small vision encoders (e.g., 93M SigLIP-B/16).
2. Extend Context Length: Increasing RoPE to 273K and fine-tuning unlocks 8K–16K token limits.
3. Aggressive Token Compression: Pixel shuffle ($r=4$) achieves a critical 16× reduction in visual tokens.
4. Sub-1GB Inference is Achievable: SmolVLM-256M operates on 0.8GB RAM, enabling edge and mobile ML.
5. Fully Open-Source: Models, datasets, code, and the HuggingSnap mobile app are open to accelerate research.